Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/26/2024
PI Name:	Dudzinski, Stephanie		
Project Title:	Extend Healthy Life in Space by Characterizing Radiation Induced Pro-Inflammatory Response and Enhancing Wound Repair and Recovery with Radiation Mitigating Thrombin Peptide		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2023 TRISH-RFA-2301: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
Start Date:	12/01/2023	**End Date:**	11/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Frank, Steven (MENTOR: University Of Texas M D Anderson Cancer Center)		
Grant/Contract No.:	NNX16AO69A-P0802		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP When astronauts exit Earth's atmosphere, they are exposed to space radiation. While a single day of exposure to space radiation levels would unlikely result in significant harm to a person, astronauts spend up to 6 months at a time on missions in space. Additionally, the time expected for Mars or extended Moon missions are projected to be up to 21 months, which is 3.5 times longer than current missions. In these settings, low levels of space radiation accumulate during long missions and can negatively impact astronauts' bodies and health in several ways, including increasing their risk for degenerative diseases, central nervous system diseases, and systemic immune dysfunction. Given that astronauts are experiencing detrimental side effects with current lengths of space travel and space radiation exposure, there is urgency to provide radioprotecting solutions. One organ significantly affected by space radiation is the eye, increasing astronauts' risk of visual deficits and cataract formation. We hypothesize that space radiation alters the immune system, which in turn increases the risk of developing ocular diseases. We plan to use mouse models treated with space radiation doses to characterize space radiation dose-induced changes to the immune microenvironment. To prevent these radiation induced diseases, we have a drug, TP508, that decreases radiation induced inflammation. We will test the ability of TP508 to prevent ocular diseases by treating the mice with eye drops containing TP508. The overall goal of our project is to characterize inflammatory changes induced by space radiation doses and to manipulate the radiation induced inflammation to prevent radiation associated diseases.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Given that it was requested for mice to be aged to 6 months prior to initiating radiation exposure, the majority of our experiments with multiple cohorts of aged mice are currently ongoing. However, we initiated a collaboration with Dr. Xiao Mao and Dr. Gregory Nelson, who completed the same fractionation with 2.08 cGy per day, 6 days a week, for 4 weeks as our experiments, but using the Galactic Cosmic Simulation, which has 33 different energetic ion beams to model the radiation fields at a central point in a human female body behind 20 g/cm2 shielding in space at solar minimum. We have received fixed eyes treating with this regiment that are undergoing multiplex immunohistochemistry staining with CD4, Foxp3, CD8, CD11b, CD206, and Phospho-VEGFR2 antibodies to analyze CD4+ helper T cells, T regulatory cells, CD8+ cytotoxic T cells, macrophages including M2-like macrophages, and vasculature growth signaling. These orbits were processed in 2 mm thick slices and mounted on a tissue-microarray for analysis. Slides were digitized. We also initiated a collaboration with Imaging Physics researchers to perform in vivo imaging of macrophages using perfluorocarbon nanoparticles with 19F MRI imaging. This will allow us to track radiation-induced macrophage recruitment in vivo during radiation treatments. These ocular MRI coils and nanoparticles have been developed and are undergoing in vivo testing currently. For the upcoming year, the preparation will accumulate in analysis of the GCR simulation treated IHC stains, treatment of our 6-month aged mice with the low dose space radiation or sham treatment using in vivo imaging of macrophages, tissue analysis of the orbits, spleen, peripheral blood, and heart to characterize the immune microenvironment, and measuring the effects of TP508 via intraorbital drops or intravenous injections as a radioprotectant and on the immune microenvironment.
Bibliography Type:	Description: (Last Updated: 03/12/2025)
Awards	Dudzinski S. "Radiological Society of North American Resident Research Grant, July 2024." Jul-2024
Awards	Frank SJ. "Quantitative Imaging Analysis Core, May 2024." May-2024